PRINCIPLES OF ECONOMICS

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FPEC 101 – LECTURE 13

Unit 2

PRINCIPLES OF ECONOMICS

Production Costs and Market Structures

FPEC 101 – LECTURE 13

Unit 2

Wage determination

Outline

- Key concepts
- Labor Productivity
- Wage Differentials
- Minimum Wage Rate
- Pay for Performance

Wage Determination

- Wages are the price that employers pay for labor
- Monetary and non monetary
- Hourly wage rate is the price paid per unit of labor services, in an hour of work.
- Labor earnings is the wage rate multiplied by the number of hours worked
- Nominal wage or money wage is the amount of money received per hour, day, or year.
- Real wage is the quantity of goods and services a worker can obtain with nominal wages; real wages reveal the “purchasing power” of nominal wages, adjusted for inflation or price rise
- Derived demand – depends on Productivity of labor and Price of the good it helps to produce
- In the long run, if productivity is increased substantially causing demand for labor to increase more rapidly than increase in the supply of labor S, it results in increased real wages.

Labor productivity

- Higher the productivity of labor, higher the demand for it.
- If the total supply of labor is fixed, then the stronger the demand for labor, the higher is the average level of real wages
- Factors affecting Labor Productivity
  - Capital availability
  - Access to abundant natural resources
  - Advanced technology More capital, and more advanced capital. Better work methods
  - Labor quality The health, vigor, education, and training
  - Other factors
    (a) the efficiency and flexibility of management
    (b) a business, social, and political environment that emphasizes production and productivity
    (c) the vast size of the domestic market, which enables firms to engage in mass production
    (d) the increased specialization of production enabled by free-trade agreements with other nations.

Purely competitive labor market

- Numerous firms compete with one another in hiring a specific type of labor.
- Each of many qualified workers with identical skills supplies that type of labor.
- Individual firms and individual workers are “wage takers,” since neither can exert any control over the market wage rate.

- Depends on
  - Market Demand for Labor
  - Market Supply of Labor

Purely competitive labor market

Market Demand for Labor

- Labor Demand curves are called the MRP, the marginal revenue product curves
- Downward sloping, higher the wage rate, lower the demand for labor
- In the theory of competitive labor markets, the demand curve for labor comes from the estimated marginal revenue product of labor (MRP_L)
- It is defined as the extra revenue generated by a firm when it employs an additional unit of labor
- MRP_L = Marginal product of labor x marginal revenue
- Marginal Revenue Product for Labor = $\frac{\text{Change in total revenue}}{\text{Unit change in factor quantity}}$

The Demand for Labour

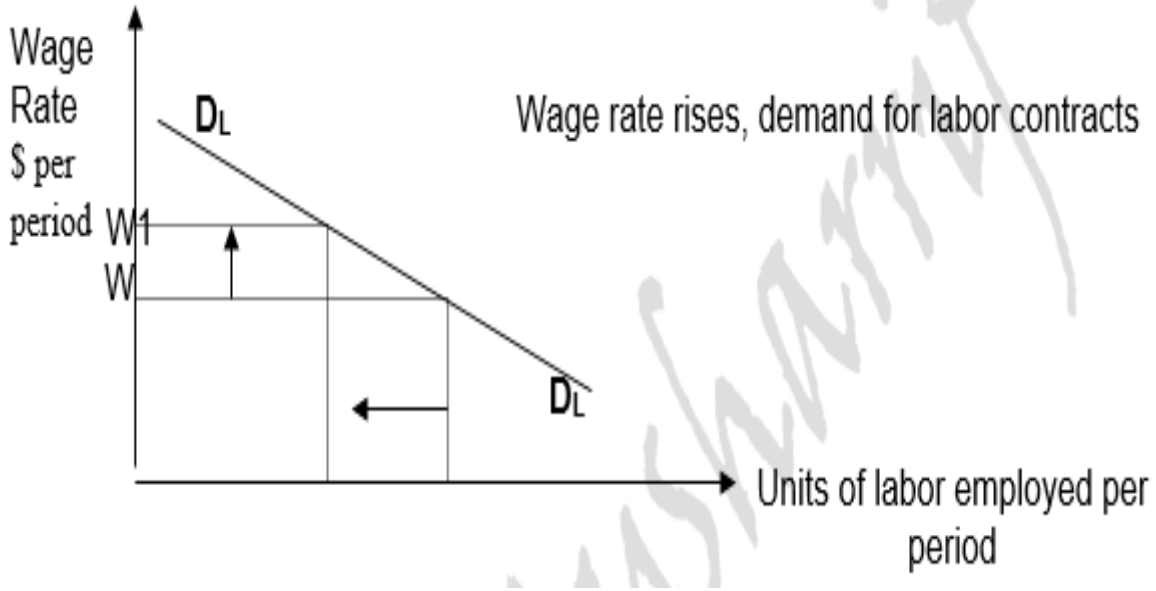
Units of a factor	Total product	Marginal product, MP	Product price	Total revenue	Marginal revenue product, MRP
0	0		$2	$ 0	
		7			$ 14
1	7		2	14	
		6			12
2	13		2	26	
		5			10
3	18		2	36	
		4			8
4	22		2	44	
		3			6
5	25		2	50	
		2			4
6	27		2	54	
		1			2
7	28		2	56	

MRP_L = Marginal product of labor x marginal revenue

Marginal Revenue Product for Labor = $\frac{\text{Change in total revenue}}{\text{Unit change in factor quantity}}$

Purely competitive labor market

Market Demand for Labor



Purely competitive labor market

Market Demand for Labor

The changing patterns of demand of Labor

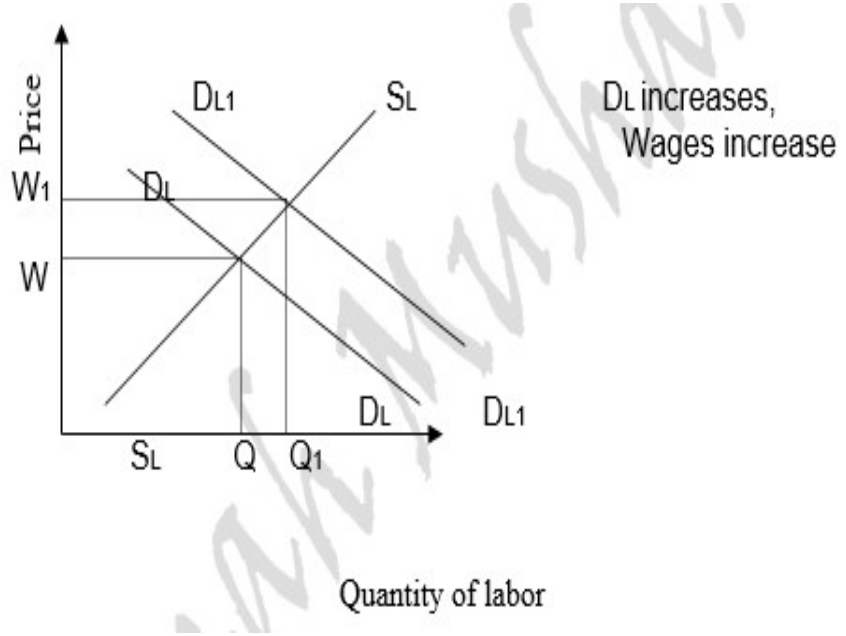
- A change in the demand for a product
- Mass production
- Division of labor to achieve economies of scale
- Worker productivity
- Changes in the price and productivity of capital
- Wages in other employment costs such as insurance, taxes etc

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Purely competitive labor market

Market Demand for Labor

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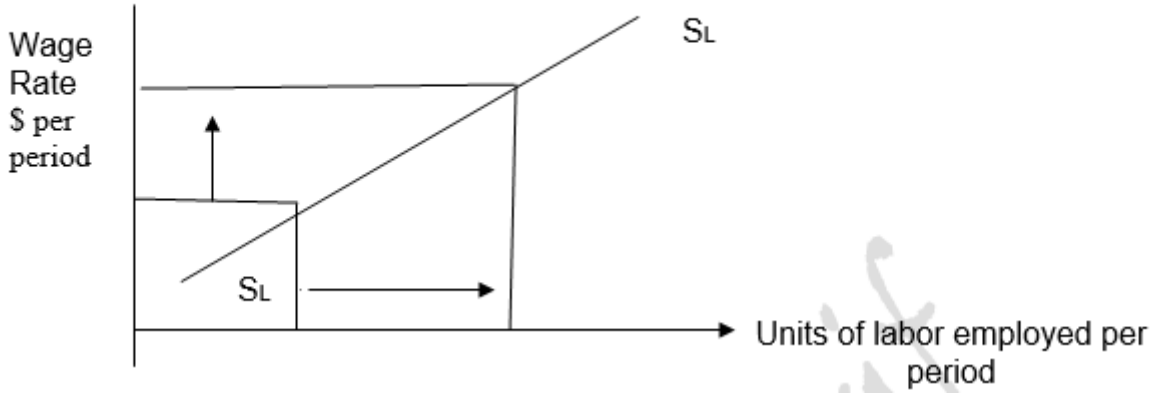
Purely competitive labor market

Market Supply of Labor

- Upward sloping, higher the wage rate, higher the supply of labor
- Higher wage rates are paid to attract workers away from alternative job opportunities available to them.
- Also to induce people who are not currently in the labor force to seek employment

Purely competitive labor market

- Market Supply of Labor



Purely competitive labor market

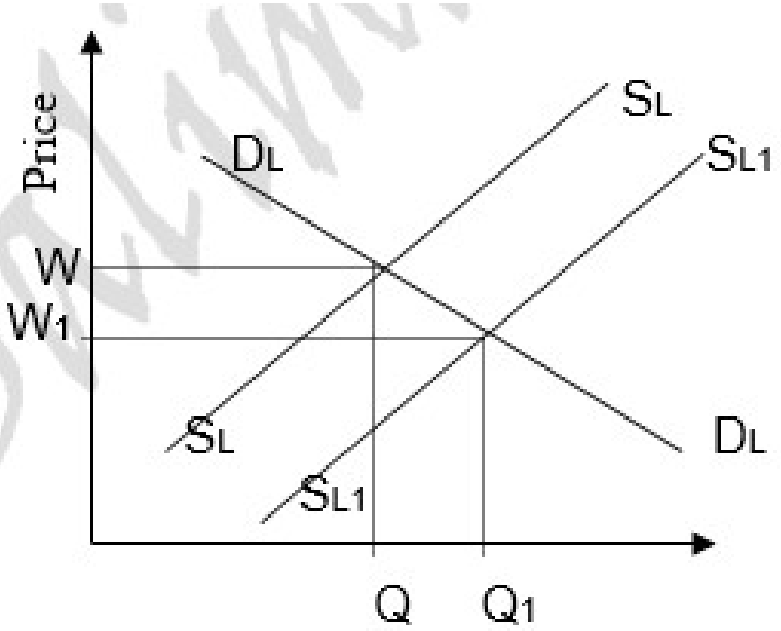
Market Supply of Labor

The changing patterns of supply of labor

- Population
- Net migration
- It is less if the job requires special training, skills
- The nature of the job
- Work environment
- Timings of work
- Education and training
- The number of people going into a certain profession
- The barriers to the entry in an occupation
- Wages in substitute occupation
- non wage factors or fringe benefits are also known as net advantages

Purely competitive labor market

Market Supply of Labor



SL increases
Wages decrease

Purely competitive labor market

Equilibrium

- Profit maximizing level of employment
- Firm considers
  - The Marginal revenue product for Labor
  - Marginal resource Cost or cost of employment
- Firm should employ labor up to the point where the Marginal Revenue Product of Labor = Marginal Cost of labor
- The cost of hiring an additional worker equals to the extra sales revenue from selling their output

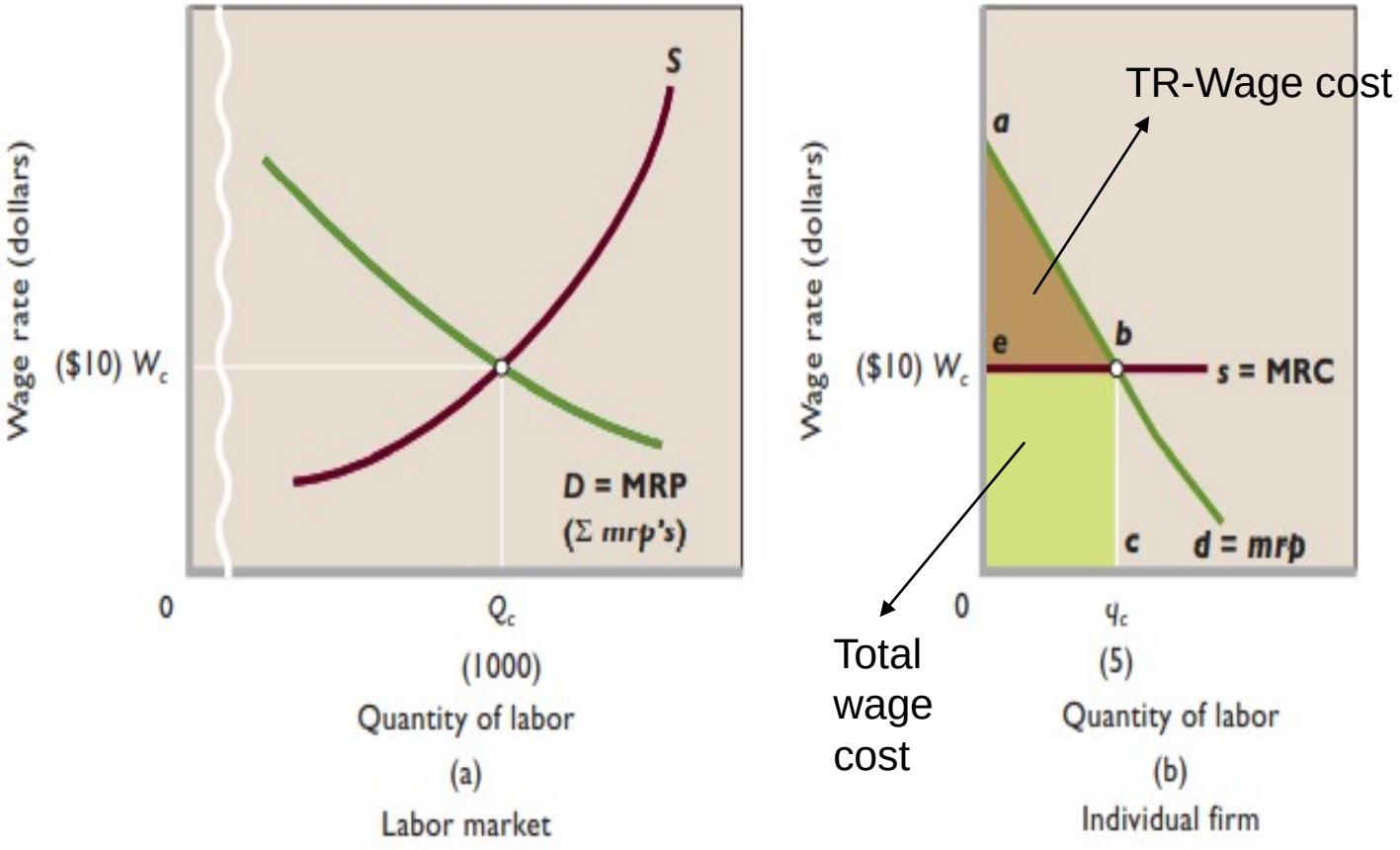
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Equilibrium

- Each of the many firms employs such a small fraction of the total available supply of this type of labor that none of them can influence the wage rate. The supply of this labor is perfectly elastic to the individual firm, as shown by horizontal line
- When the price of a resource is imposed on the individual competitive firm, the marginal cost of that resource (MRC) is constant and is equal to the resource price. Here, MRC is constant and is equal to the wage rate. Each additional worker hired adds precisely his or her own wage rate to the firm's total resource cost. So the firm in a purely competitive labor market maximizes its profit by hiring workers to the point at which its wage rate equals MRP or MRC = MRP

Purely competitive labor market

Equilibrium for market and firm (profit maximizing level)



- The equilibrium wage rate Wc and the number of workers Qc in (b)are determined by labor S and D. (a) Because this *market wage rate* is given to *the individual firm* hiring in this market, its labor supply curves MRC is perfectly elastic.
- Its labor demand curve, d, is its MRP curve
- Firm maximizes its profit by hiring workers up to MRP = MRC.
- Area 0abc represents both the firm's total revenue and its total cost.
- The green area is its total wage cost
- The brown area is its nonlabor costs, including a normal profit— that is, the firm's payments to the suppliers of land, capital, and entrepreneurship.
- Normal profit earned by firm

Mc Connell & Brue

Purely competitive labor market

Elasticity of Market Demand for Labor

- Responsiveness of labor demand due to a change in the wage rate
- Depends on
  - Labor cost as percentage of total costs
  - Ease and cost of factor substitution
  - Price elasticity of demand for final output

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Purely competitive labor market

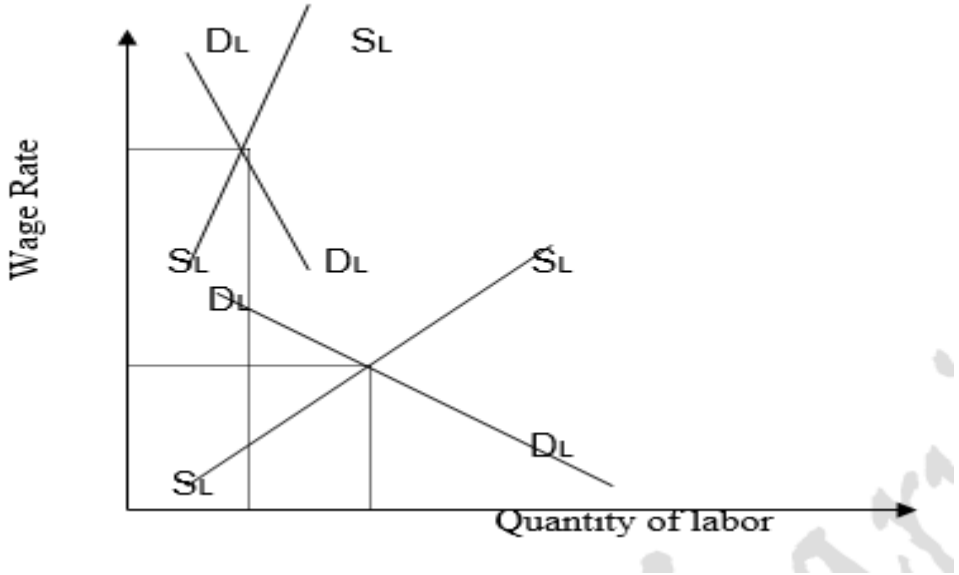
Elasticity of Market Supply for Labor

- Responsiveness of labor supply due to a change in the wage rate
- Depends on
  - Availability of Good Substitutes to specific labor
  - Specific skills and educational requirements
  - Time period
    - Short run, inelastic as it takes time for people to respond to changes in relative wages.

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Purely competitive labor market

- For skilled labor, demand and supply is relatively inelastic; therefore, there is a higher wage lesser supply of labor available.
- For unskilled workers, the demand and supply for labor is elastic because there are fewer skills required and more labor supply of the available



Wage Differentials

Differences in wages between different occupations

- Position in an occupation
- Qualifications and experience
- Education
- Racial or gender discrimination
- Skills
- Nature of the job …risky or timings
- Whether it's in the private sector or public sector
- Employer discrimination
- Trade Unions

Wage Differentials

Differences in wages between different occupations

- Immobility
- Satisfaction
- Efficiency and hard work
- Profession you are involved in, example lawyers and doctors get high pay
- The reputation of the institute one has studied from
- Currency differences
- Non monetary fringe benefits
- The mobility of workers… if this is low, to increase mobility benefits such as housing Etc. may be given to encourage workers to change their profession.

Minimum Wage Rate

- A “wage floor” that will help less skilled workers earn enough income to escape poverty.
- Impact

  - Rising costs of production

  -May lead to decrease in workers hired

  -Poorly targeted to reduce poverty

  +Job losses may not occur in markets with short labor supply and great labor demand

  +May increase labor supply

  +Increases labor productivity
- Conclusion
  - More workers are helped by the minimum wage than are hurt
  - Gives society some assurance that employers are not “taking undue advantage” of vulnerable, low-skilled worker

Pay for performance

- Piece Rates- compensation paid according to the number of units of output a worker produce
- Commissions -Employees who sell products or services
- Royalties -certain percentage of sales revenues from their works
- Bonuses- payments in addition to one's annual salary
- Stock options- buy shares of their employer's stock at a fixed, lower price when the stock price rises
- Profit Sharing- allocate a percentage of a firm's profit to its employees.
- Efficiency Wages- above-equilibrium wage

SUMMARY

- Nominal vs real wage
- Role of Productivity
- Purely competitive labor market
  - Demand
  - Supply
  - Profit maximizing equilibrium
- Wage Differentials
- Minimum Wage Rate
- Pay for Performance